Programming

1. Basic Requirements

The following are supplementary requirements for Programming.

Applications

- Git
- Visual Studio Code
- Android Studio

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2. Environment Setup

The following environment must be configured before you begin. If you have any questions, please consult an administrator.

Web Browser

Bookmark the following additional websites:

Name	URL
Programming Resources	https://www.firstinspires.org/resources/library/ftc/programming-resources
REV Robotics Documentation	https://docs.revrobotics.com/duo-control/hello-robot-java/welcome

Name	URL
Pedro Pathing	https://pedropathing.com/
Robot Dashboard	http://192.168.43.1:8080/

Git

Git is the version control tool adopted by our team for program files. It is installed automatically along with Android Studio.

Android Studio

Android Studio is the IDE adopted by our team for writing programs.

Software installation and configuration:

1. Open https://developer.android.com/studio
2. Click "Download Android Studio" and install the version for your platform.

On Windows, note the following during installation:

- Both checkboxes must be checked
- Choose a path with sufficient space that will not be moved

Initialization:

- Select "Standard" mode
- Check "Accept" when agreeing to the licenses
- Keep everything else unchanged and click "Next"

Installing the Chinese language pack (optional):

1. Open the Android Studio Chinese Language Pack releases
2. Download the latest language pack (.jar file)
3. In the left tab list on the welcome screen, select "Plugins", then "Install Plugin from Disk"
4. Select the downloaded .jar file and open it; make sure the plugin is enabled once loaded

5. In the left tab list, select "Customize", choose "Americas" under "Regions", then select "Chinese (Simplified)" under "Language" and restart

Cloning the repository:

1. Click "GitHub" in the left tab list and sign in via "Log in with GitHub" to authorize.
2. Select the current season's code repository (e.g., `ftc32477/FTC-32477-Decode-Program`)
3. Choose an empty folder on a path that will not change and click "Clone"
4. Wait for the download to finish; track progress via the "Build" tab in the left sidebar or the progress bar in the bottom-right corner

> If you run into problems, ask an administrator.

Visual Studio Code

Visual Studio Code is the tool adopted by our team for viewing code history.

Software installation and configuration:

1. Open https://code.visualstudio.com/Download
2. Download and install the build for your platform
3. Install the Chinese language pack if you want the UI in Chinese.

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3. Tools Overview

Android Studio

Official documentation: https://developer.android.com/studio/intro?hl=en

In this project, we build on the official FTC application framework and write robot control programs in Java under the `TeamCode` folder, calling upon and simulating the various dependency library environments needed for robot operation.

For the essentials you need to know about Android Studio, refer to the quick UI walkthrough in the official documentation.

Visual Studio Code

Official documentation: https://code.visualstudio.com/docs

Since Visual Studio Code follows similar operating logic to Android Studio and is used less frequently in this project, refer to the Android Studio section for interface guidance.

Robot Dashboard

- Connect to the Wi-Fi network "`32477-RC`". Wi-Fi password: ask an administrator or obtain it via the Driver Hub.
- URL: http://192.168.43.1:8080/
- This is the web page of the Wi-Fi module built into the Control Hub, providing a graphical management console for the Control Hub.

For the essentials you need to know about the Robot Dashboard, refer to the quick UI walkthrough in the official documentation.

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4. Workflow

The Programming team's core work consists of three parts:

- **Autonomous programs**: control logic for the season's Autonomous period
- **TeleOp programs**: operation logic for the driver-controlled period
- **Sensor configuration**: configuration of sensors and vision systems

Debugging Is the Core

> The real difficulty in program development lies in debugging. Autonomous paths, driver controls, PID parameters, vision configuration — most sensors come with ready-made packages you can reuse. What you really do is *tune*.

Debugging runs through the entire development workflow:

1. **Requirements analysis**: clarify the rules and task requirements of the current season
2. **Architecture design**: design the overall program architecture and module breakdown
3. **Implementation**: write Java code in Android Studio
4. **Version control**: manage code versions with Git
5. **Debugging**: tune autonomous paths, driver controls, PID parameters, and vision configuration
6. **Testing & verification**: verify program functionality on the robot
7. **Code review**: review and merge code via GitHub
8. **Deployment**: deploy the final build to the Robot Controller